



Balanced on the Roster, Not the Risers

validating a community choir's automated seating-balance score against audience-rated sound

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Motivation

A seating-balance tool now reassigns choir sections rehearsal to rehearsal to keep parts numerically even. Does a tidier balance score make the choir sound any different — or just look better on the roster?

What we set out to test

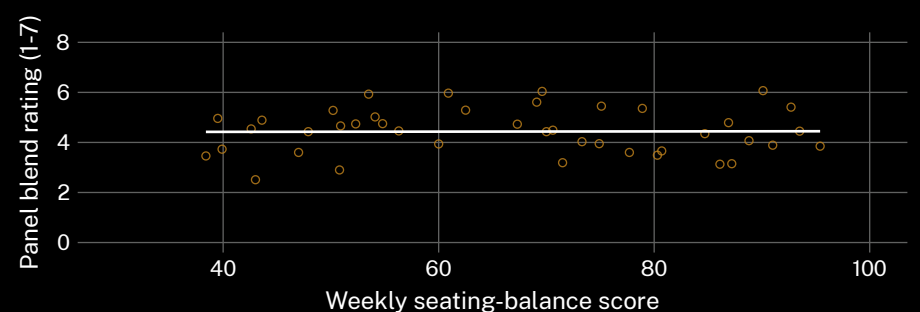
- Whether the weekly balance score predicts an independent panel's blend rating of the same rehearsal
- Whether tool-assigned seating changes how singers describe their own spot — chosen, or logged
- Whether choirs quietly override the tool's suggested seating as a concert date nears

Study design

- 18 community choirs (9 tool-assigned, 9 self-selected) · one 14-week season
- blind 6-rater panel scored a random sample of 42 of 240 logged weekly recordings, 7-point blend scale
- balance-score log and seating overrides pulled from the tool's own audit trail, not self-report

Findings

The balance score climbed steadily across the season at tool-assigned choirs. Panel-rated blend did not track it ($r = 0.02$, $n = 42$). Tool-assigned singers were more than twice as likely to describe their spot as “where I was put” than singers who chose their own.



Across 42 blind-adjudicated rehearsal recordings from March–June 2026, the weekly seating-balance score showed no reliable relationship with panel-rated blend ($r = 0.02$).

Interpretation

A balance score that improves independently of the sound it is meant to track is measuring something else — attendance, self-report compliance, or its own inputs. Section balance may be necessary for good blend; it is evidently not sufficient, and a tidy roster is no guarantee of a tidy sound. The override log tells a parallel story: overrides cluster in the fortnight before a concert, when directors reach for the seating they already trust. Next: audit whether singers' self-reported vocal range drifts toward whatever the tool needs to balance.

Further reading

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3. Ternström, S. (1991). Physical and acoustic factors that interact with the singer to produce the choral sound. *Journal of Voice*, 5(2), 128–143. doi:10.1016/s0892-1997(05)80177-8
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5. Strathern, M. (1997). 'Improving ratings': audit in the British University system. *European Review*, 5(3), 305–321. doi:10.1002/(SICI)1234-981X(199707)5:3<305::AID-EUR0184>3.0.CO;2-4

We thank the choirs and adjudication panel of the instrumented rehearsal season. Recordings de-identified before adjudication; panel blind to seating condition; no singer's individual rating retained.

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